

Technique Recursive sum of powers

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1 Technique Recursive sum of powers

To solve for $\sum_{i=1}^n i^p$ consider $\sum_{i=1}^n (1+i)^{p+1} - i^{p+1} = (1+n)^{p+1} - 1$

1.1 Sum of Squares

Let $p=2$

$$\sum_{i=1}^n (1+i)^3 - i^3 = \sum_{i=1}^n 1 + 3i + 3i^2 + i^3 - i^3 = \sum_{i=1}^n 1 + 3i + 3i^2$$

and

$$\sum_{i=1}^n (1+i)^3 - i^3 = (n+1)^3 - 1$$

Hence

$$\sum_{i=1}^n 1 + 3 \sum_{i=1}^n i + 3 \sum_{i=1}^n i^2 = (n+1)^3 - 1$$

$$n + 3 \frac{n(n-1)}{2} + 3 \sum_{i=1}^n i^2 = (n+1)^3 - 1$$

$$3 \sum_{i=1}^n i^2 = n^3 + 3n^2 + 3n + 1 - n - 3 \frac{n(n-1)}{2}$$

$$3 \sum_{i=1}^n i^2 = n(n+1)\left(n + \frac{1}{2}\right)$$

$$\boxed{\sum_{i=1}^n i^2 = \frac{n}{3} \cdot (n+1) \cdot \left(n + \frac{1}{2}\right)}$$

1.2 Sum of Cubes

Let $p=3$

$$\sum_{i=1}^n (1+i)^4 - i^4 = \sum_{i=1}^n 1 + 4i + 6i^2 + 4i^3 + i^4 - i^4 = \sum_{i=1}^n 1 + 4i + 6i^2 + 4i^3$$

and

$$\sum_{i=1}^n (1+i)^4 - i^4 = (n+1)^4 - 1$$

Hence

$$\sum_{i=1}^n 1 + 4 \sum_{i=1}^n i + 6 \sum_{i=1}^n i^2 + 4 \sum_{i=1}^n i^3 = (n+1)^4 - 1$$

$$n + 4 \frac{n(n-1)}{2} + 6 \frac{n}{3} \cdot (n+1) \cdot (n + \frac{1}{2}) + 4 \sum_{i=1}^n i^3 = (n+1)^4 - 1$$

$$4 \sum_{i=1}^n i^3 = n^4 + 4n^3 + 6n^2 + 4n - n - 2n(n-1) - n \cdot (n+1) \cdot (2n+1)$$

Simplifying

$$4 \sum_{i=1}^n i^3 = n^4 + 2n^3 + n^2$$

$$4 \sum_{i=1}^n i^3 = n^2(n^2 + 2n + 1) = n^2(n+1)^2$$

$$\boxed{\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2}$$